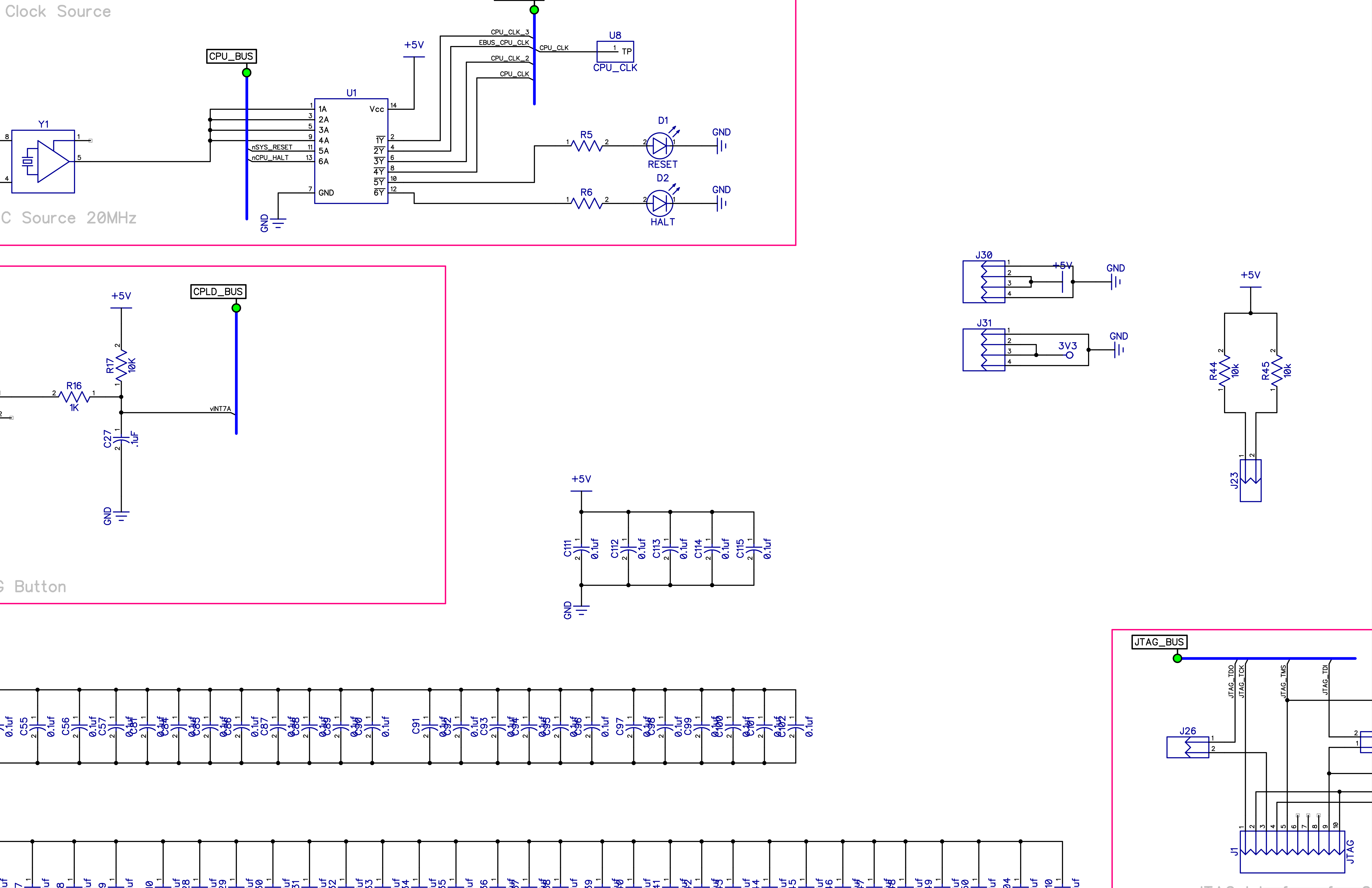
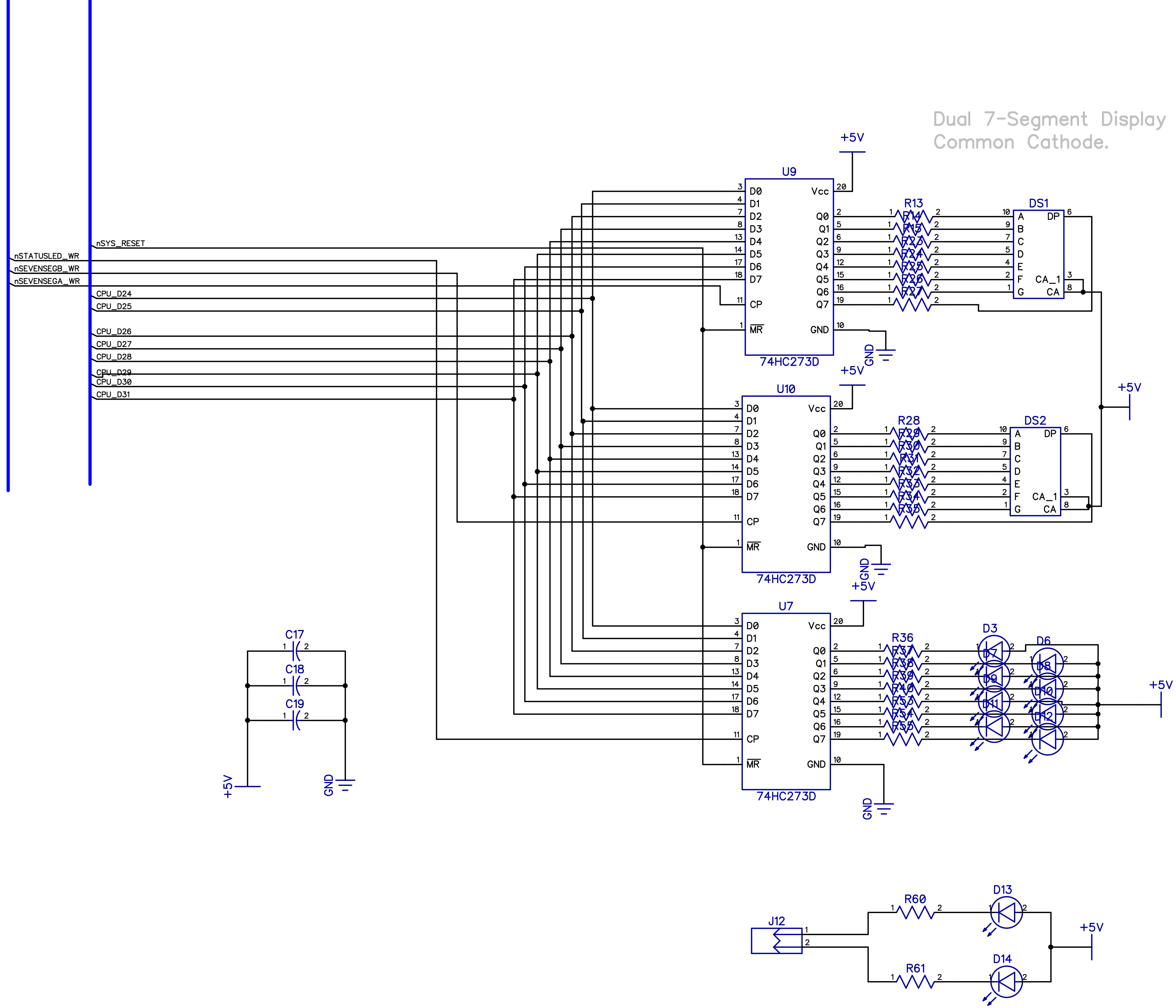
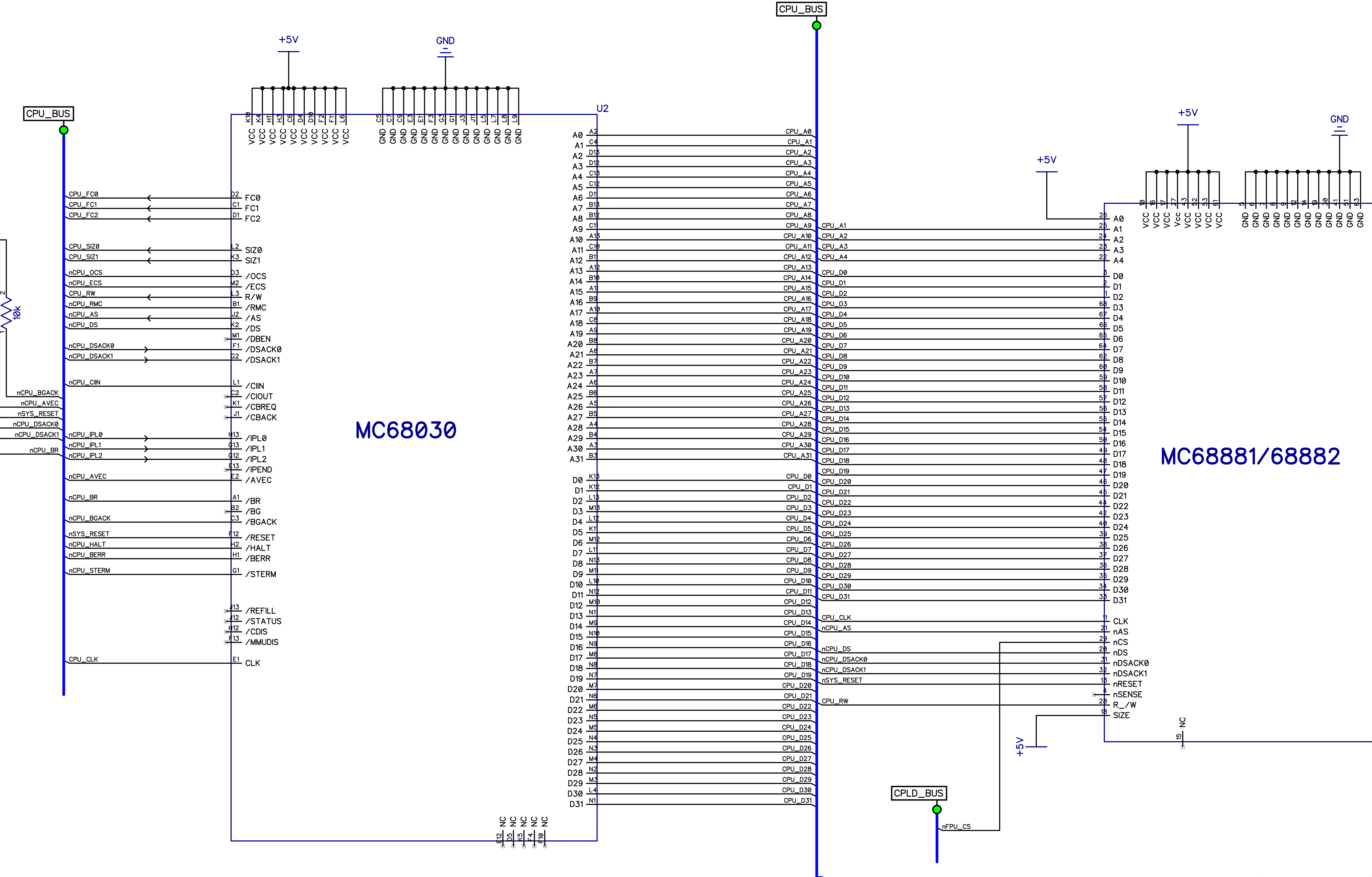


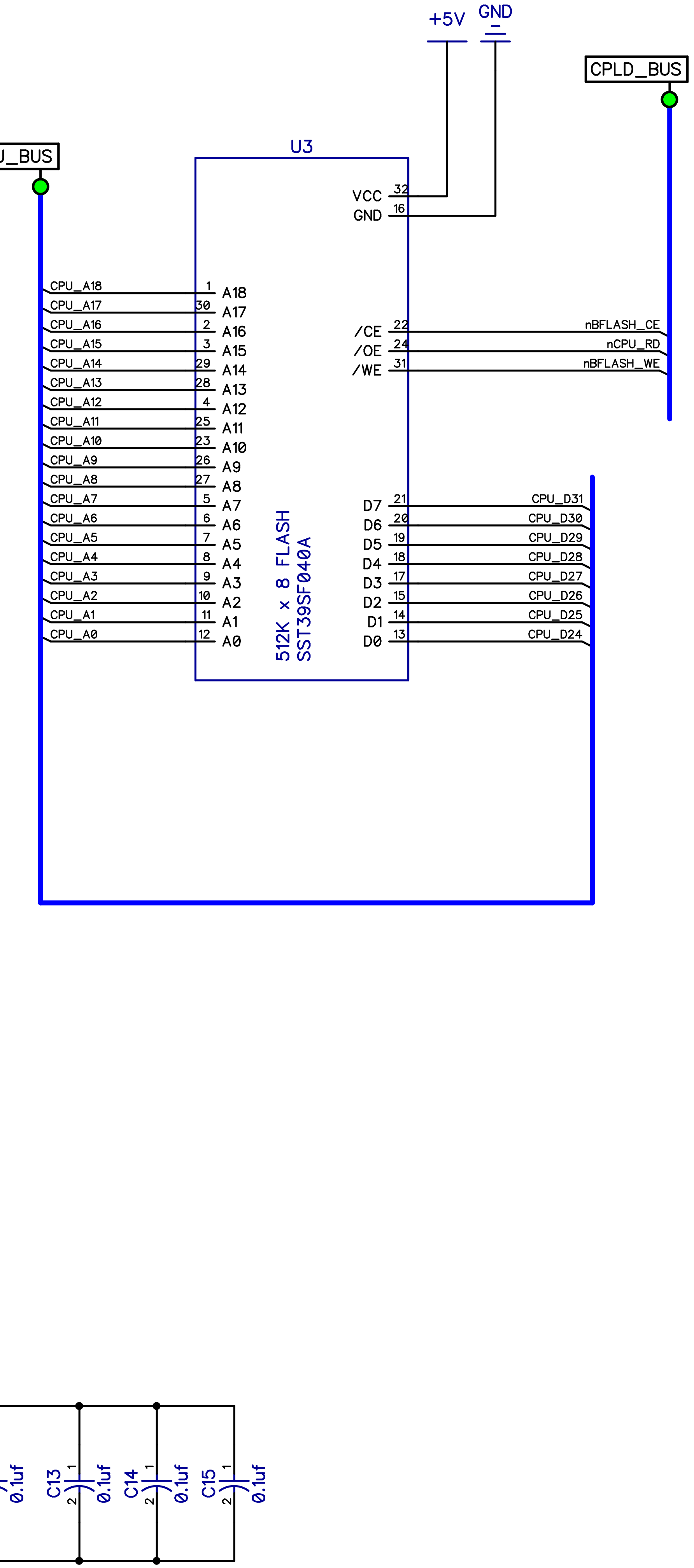
Clock Source
CPU_BUS
U1
1A
2A
3A
4A
5A
6A
Vcc
GND
1Y
2Y
3Y
4Y
5Y
6Y
nSYS_RESET
nCPU_HALT
CPU_CLK_3
EBUS_CPU_CLK
CPU_CLK_2
CPU_CLK
U8
TP
CPU_CLK
+5V
Y1
R5
R6
D1
D2
RESET
HALT
GND
C Source 20MHz
CPLD_BUS
R17
10K
R16
1K
vINT7A
C27
.1uF
G Button
J30
J31
3V3
R44
10k
R45
10k
J23
C111
C112
C113
C114
C115
0.1uf
JTAG_BUS
JTAG_TDO
JTAG_TCK
JTAG_TMS
JTAG_TDI
J26
J1
JTAG
C55
C56
C57
C87
C91
C93
C97
C99



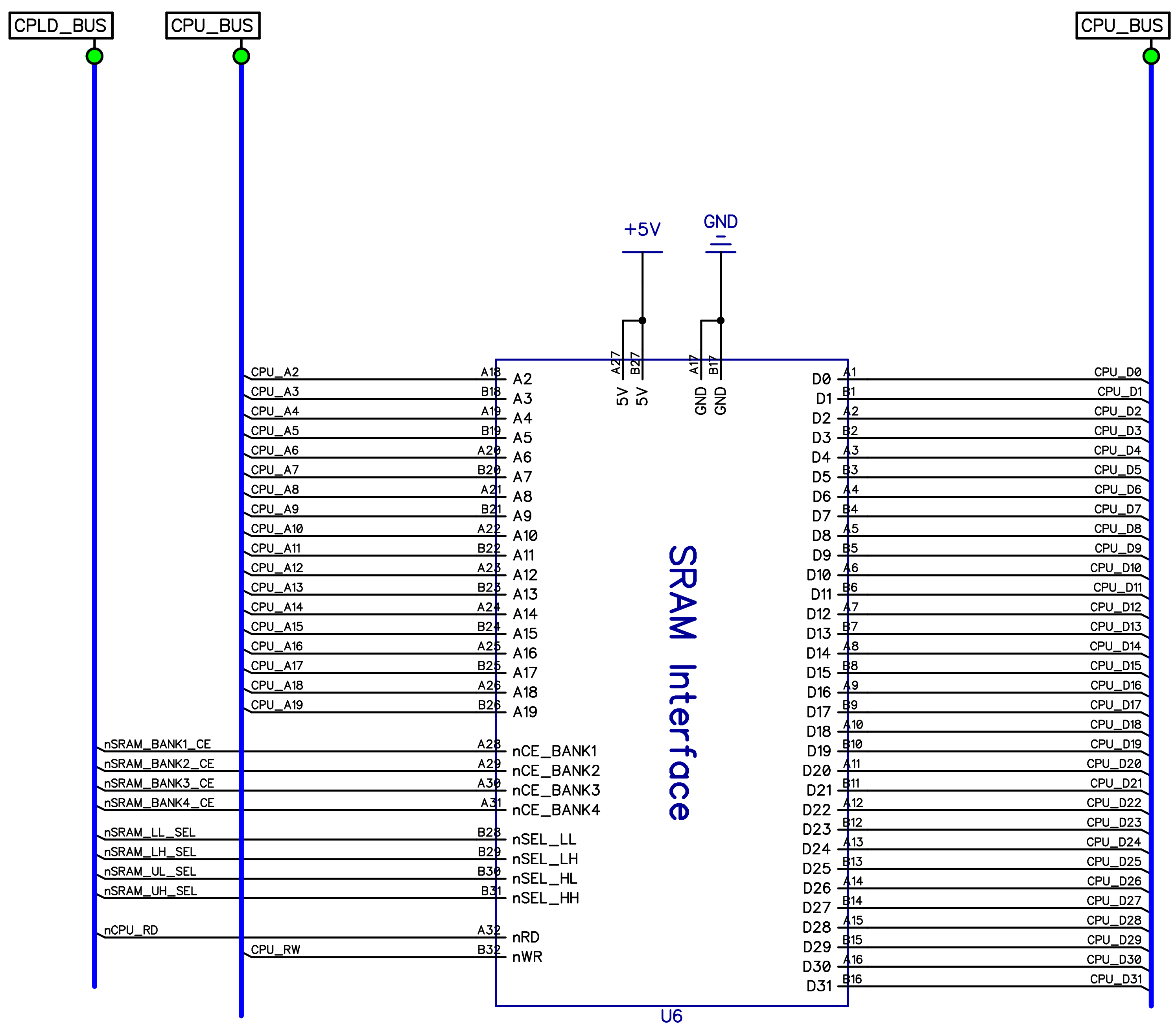
Dual 7-Segment Display
Common Cathode.
nSYS_RESET
nSTATUSLED_WR
nSEVENSEGB_WR
nSEVENSEGA_WR
CPU_D24
CPU_D25
CPU_D26
CPU_D27
CPU_D28
CPU_D29
CPU_D30
CPU_D31
U9
U10
U7
74HC273D
D0
D1
D2
D3
D4
D5
D6
D7
CP
MR
Vcc
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7
GND
+5V
DS1
DS2
DP
CA_1
CA
R13
R28
R36
R60
R61
D3
D6
D13
D14
J12
C17
C18
C19



CPU_BUS
+5V
GND
U2
MC68030
MC68881/68882
CPLD_BUS
nFPU_CS
10k
FC0
FC1
FC2
SIZ0
SIZ1
/OCS
/ECS
R/W
/RMC
/AS
/DS
/DBEN
/DSACK0
/DSACK1
/CIIN
/CIOUT
/CBREQ
/CBACK
/IPL0
/IPL1
/IPL2
/IPEND
/AVEC
/BR
/BG
/BGACK
/RESET
/HALT
/BERR
/STERM
/REFILL
/STATUS
/CDIS
/MMUDIS
CLK
nAS
nCS
nDS
nDSACK0
nDSACK1
nRESET
nSENSE
R_/W
SIZE
NC



+5V
GND
CPLD_BUS
U_BUS
U3
VCC
GND
32
16
CPU_A18
CPU_A17
CPU_A16
CPU_A15
CPU_A14
CPU_A13
CPU_A12
CPU_A11
CPU_A10
CPU_A9
CPU_A8
CPU_A7
CPU_A6
CPU_A5
CPU_A4
CPU_A3
CPU_A2
CPU_A1
CPU_A0
A18
A17
A16
A15
A14
A13
A12
A11
A10
A9
A8
A7
A6
A5
A4
A3
A2
A1
A0
/CE
/OE
/WE
nBFLASH_CE
nCPU_RD
nBFLASH_WE
512K x 8 FLASH
SST39SF040A
D7
D6
D5
D4
D3
D2
D1
D0
CPU_D31
CPU_D30
CPU_D29
CPU_D28
CPU_D27
CPU_D26
CPU_D25
CPU_D24
C13
C14
C15
0.1uf
0.1uf
0.1uf



CPLD_BUS
CPU_BUS
CPU_BUS
+5V
GND
SRAM Interface
U6
CPU_A2
CPU_A3
CPU_A4
CPU_A5
CPU_A6
CPU_A7
CPU_A8
CPU_A9
CPU_A10
CPU_A11
CPU_A12
CPU_A13
CPU_A14
CPU_A15
CPU_A16
CPU_A17
CPU_A18
CPU_A19
nSRAM_BANK1_CE
nSRAM_BANK2_CE
nSRAM_BANK3_CE
nSRAM_BANK4_CE
nSRAM_LL_SEL
nSRAM_LH_SEL
nSRAM_UL_SEL
nSRAM_UH_SEL
nCPU_RD
CPU_RW
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
A16
A17
A18
A19
nCE_BANK1
nCE_BANK2
nCE_BANK3
nCE_BANK4
nSEL_LL
nSEL_LH
nSEL_HL
nSEL_HH
nRD
nWR
5V
5V
GND
GND
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
D16
D17
D18
D19
D20
D21
D22
D23
D24
D25
D26
D27
D28
D29
D30
D31
CPU_D0
CPU_D1
CPU_D2
CPU_D3
CPU_D4
CPU_D5
CPU_D6
CPU_D7
CPU_D8
CPU_D9
CPU_D10
CPU_D11
CPU_D12
CPU_D13
CPU_D14
CPU_D15
CPU_D16
CPU_D17
CPU_D18
CPU_D19
CPU_D20
CPU_D21
CPU_D22
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CPU_D24
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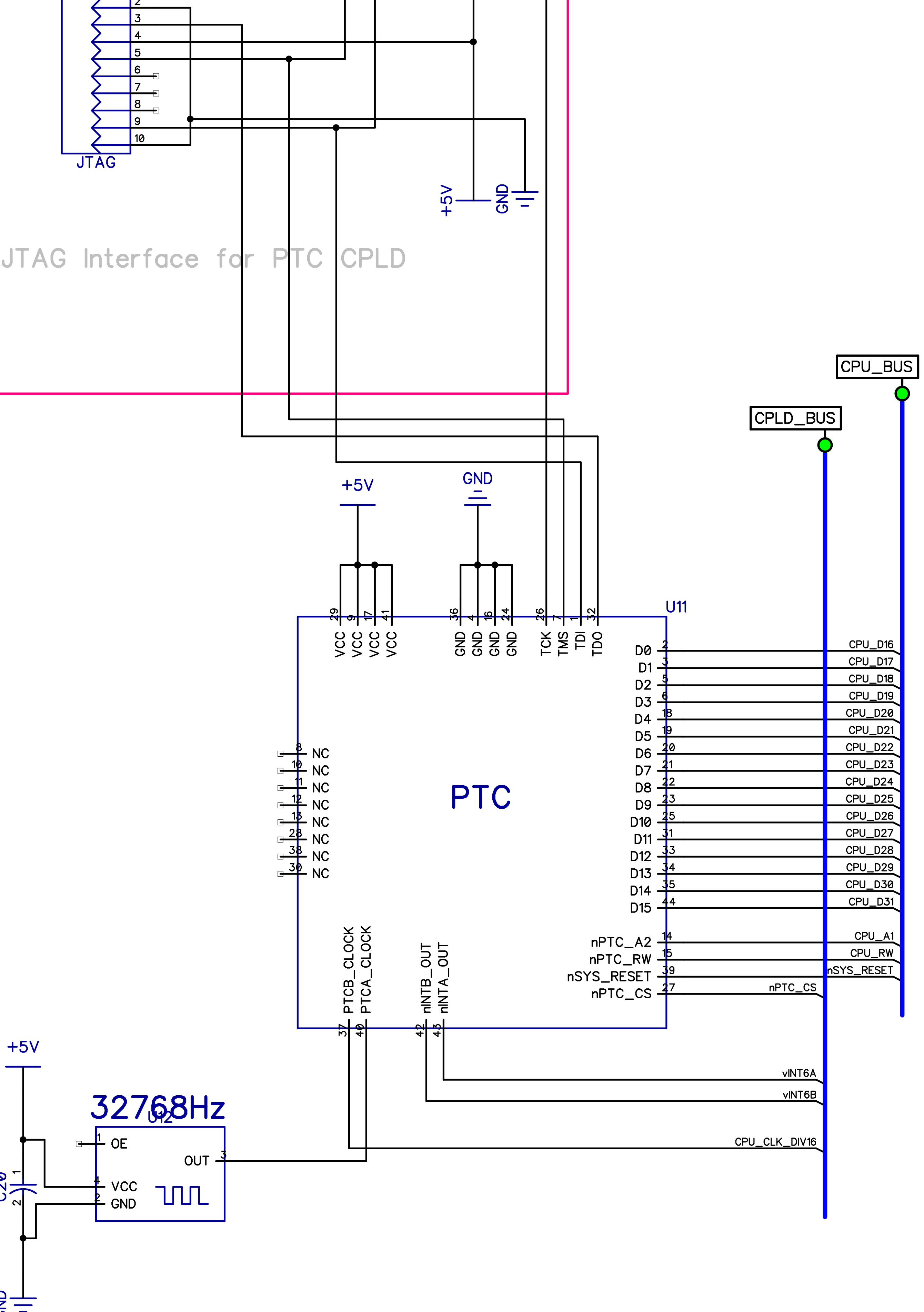
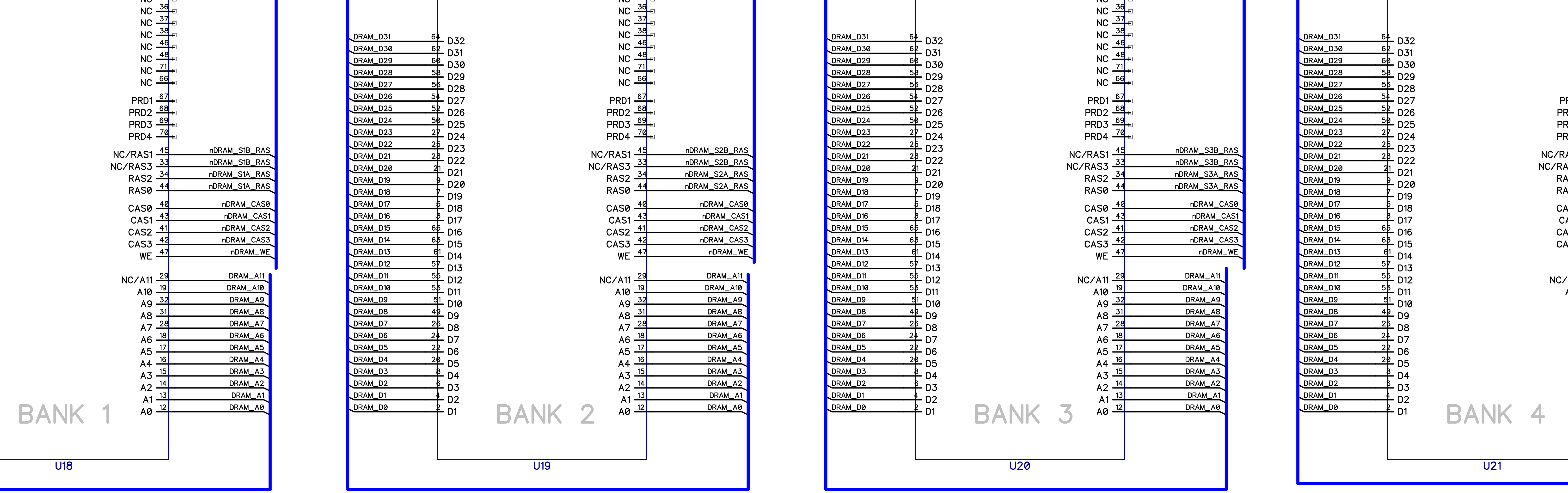
CPU_D27
CPU_D28
CPU_D29
CPU_D30
CPU_D31

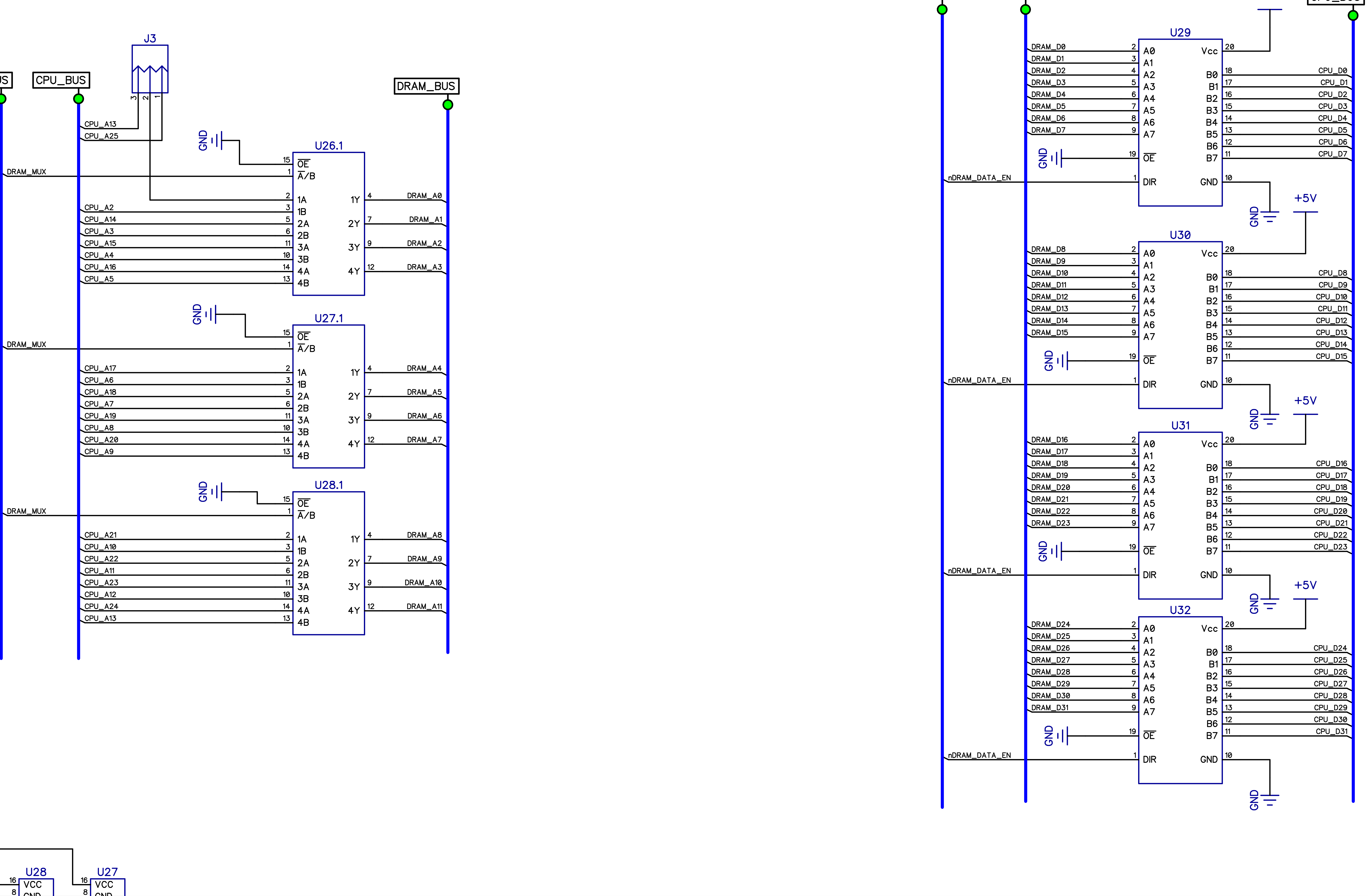

JTAG
+5V
GND
JTAG Interface for PTC CPLD
CPU_BUS
CPLD_BUS
U11
PTC
VCC
GND
TCK
TMS
TDI
TDO
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
NC
CPU_D16
CPU_D17
CPU_D18
CPU_D19
CPU_D20
CPU_D21
CPU_D22
CPU_D23
CPU_D24
CPU_D25
CPU_D26
CPU_D27
CPU_D28
CPU_D29
CPU_D30
CPU_D31
nPTC_A2
nPTC_RW
nSYS_RESET
nPTC_CS
CPU_A1
CPU_RW
nSYS_RESET
nPTC_CS
PTCB_CLOCK
PTCA_CLOCK
nINTB_OUT
nINTA_OUT
vINT6A
vINT6B
CPU_CLK_DIV16
32768Hz
U12
OE
OUT
VCC
GND



BANK 1
U18
BANK 2
U19
BANK 3
U20
BANK 4
U21
NC
PRD1
PRD2
PRD3
PRD4
NC/RAS1
NC/RAS3
RAS2
RAS0
CAS0
CAS1
CAS2
CAS3
WE
NC/A11
A10
A9
A8
A7
A6
A5
A4
A3
A2
A1
A0
nDRAM_S1B_RAS
nDRAM_S1A_RAS
nDRAM_S2B_RAS
nDRAM_S2A_RAS
nDRAM_S3B_RAS
nDRAM_S3A_RAS
nDRAM_CAS0
nDRAM_CAS1
nDRAM_CAS2
nDRAM_CAS3
nDRAM_WE
DRAM_A11
DRAM_A10
DRAM_A9
DRAM_A8
DRAM_A7
DRAM_A6
DRAM_A5
DRAM_A4
DRAM_A3
DRAM_A2
DRAM_A1
DRAM_A0
DRAM_D31
DRAM_D30
DRAM_D29
DRAM_D28
DRAM_D27
DRAM_D26
DRAM_D25
DRAM_D24
DRAM_D23
DRAM_D22
DRAM_D21
DRAM_D20
DRAM_D19
DRAM_D18
DRAM_D17
DRAM_D16
DRAM_D15
DRAM_D14
DRAM_D13
DRAM_D12
DRAM_D11
DRAM_D10
DRAM_D9
DRAM_D8
DRAM_D7
DRAM_D6
DRAM_D5
DRAM_D4
DRAM_D3
DRAM_D2
DRAM_D1
DRAM_D0
D32
D31
D30
D29
D28
D27
D26
D25
D24
D23
D22
D21
D20
D19
D18
D17
D16
D15
D14
D13
D12
D11
D10
D9
D8
D7
D6
D5
D4
D3
D2
D1

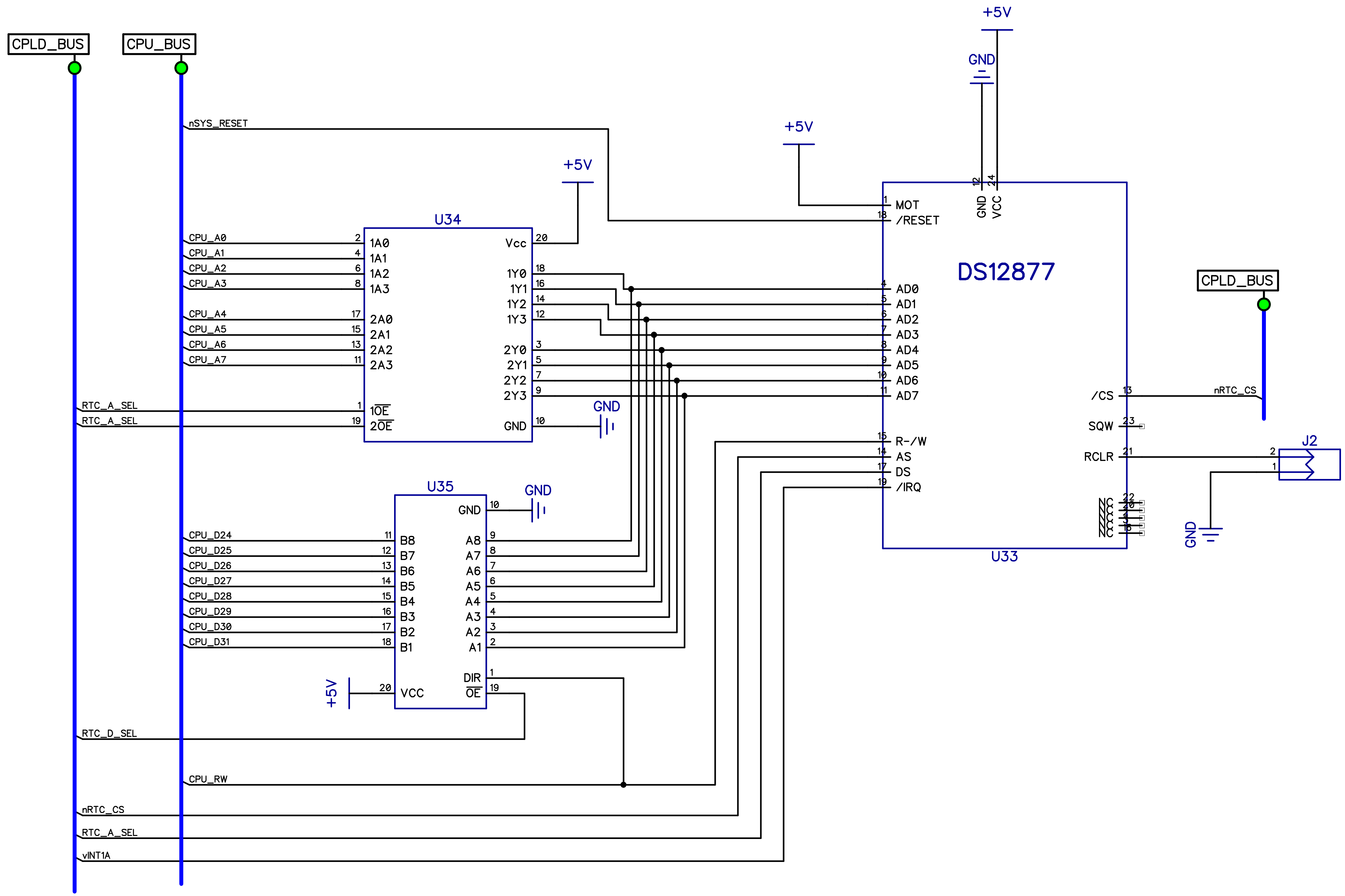
DRAM Address MUX
Jumper J3 selects between 16/32MBSIMMS[2-3] and
64/128MB SIMMS[1-2]

DRAM_MUX comes from the Bus Control CPLD

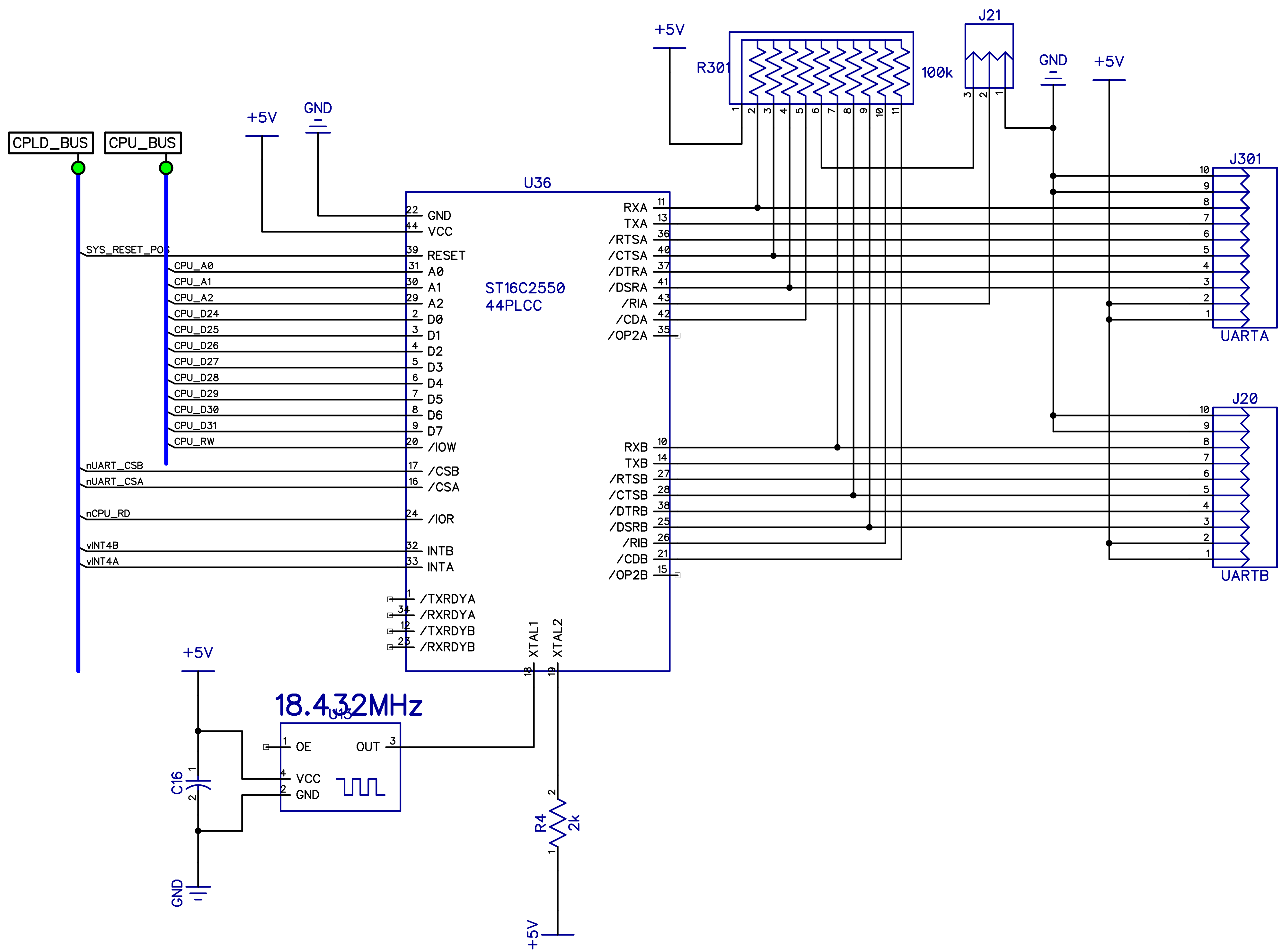



DRAM Data Trance
OutputEnable is alw
and DIR is kepts c
signal is going fro
unless a DRAM RE

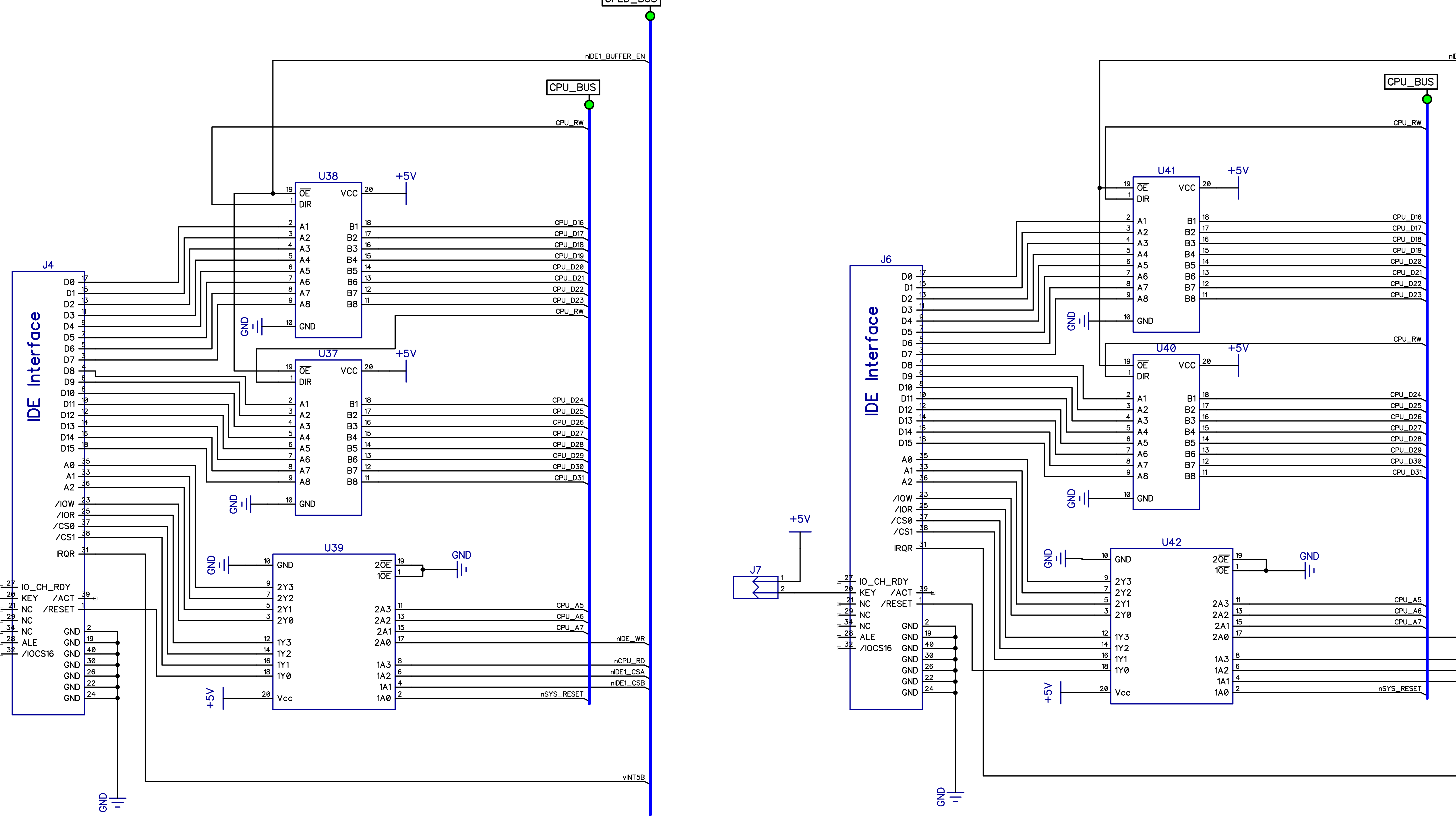
nDRAM_DATA_EN c
Control CPLD



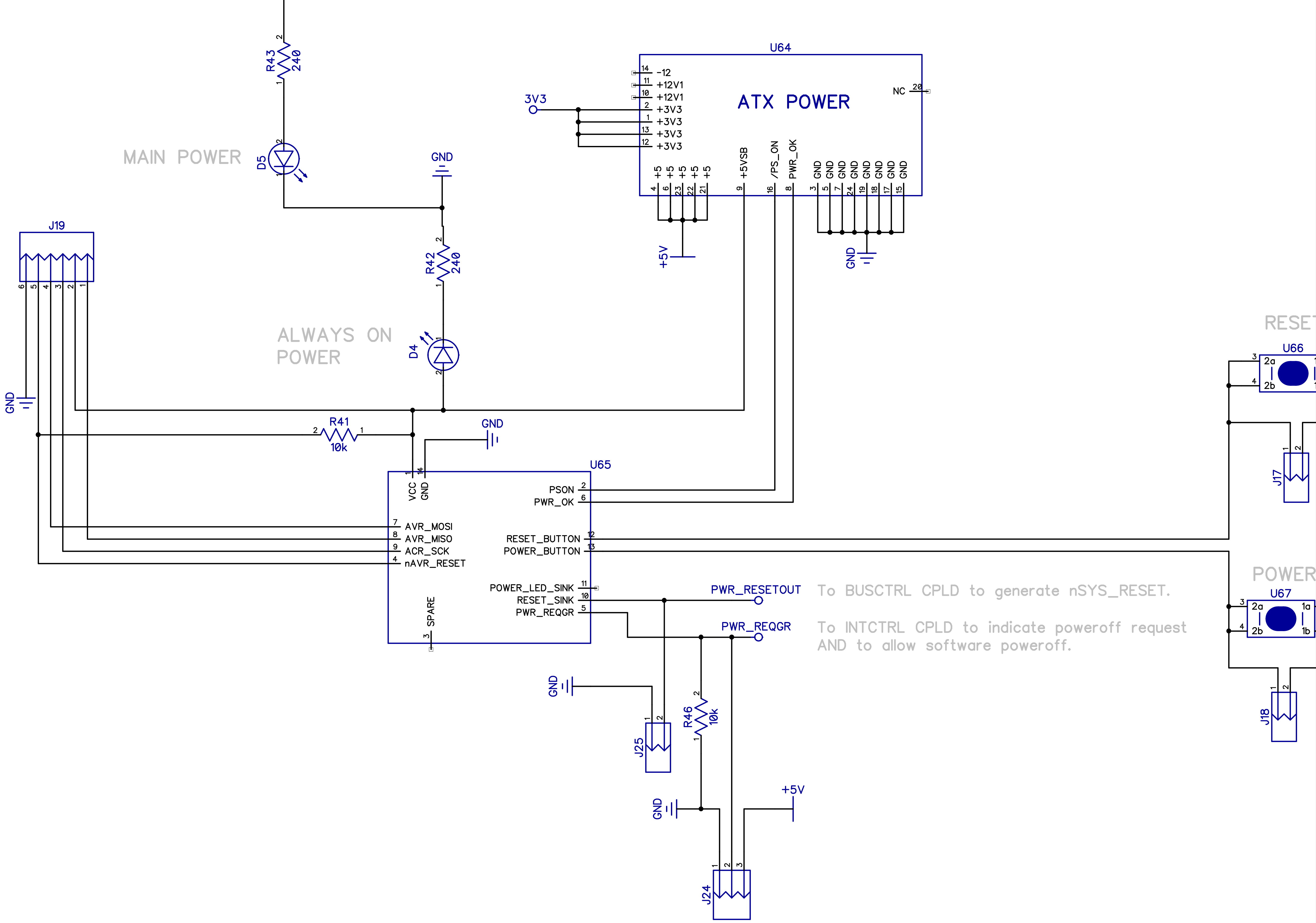
CPLD_BUS
CPU_BUS
nSYS_RESET
+5V
GND
VCC
MOT
/RESET
DS12877
AD0
AD1
AD2
AD3
AD4
AD5
AD6
AD7
/CS
SQW
RCLR
R-/W
AS
DS
/IRQ
U33
nRTC_CS
J2
U34
1A0
1A1
1A2
1A3
2A0
2A1
2A2
2A3
Vcc
1Y0
1Y1
1Y2
1Y3
2Y0
2Y1
2Y2
2Y3
1OE
2OE
CPU_A0
CPU_A1
CPU_A2
CPU_A3
CPU_A4
CPU_A5
CPU_A6
CPU_A7
RTC_A_SEL
U35
B8
B7
B6
B5
B4
B3
B2
B1
A8
A7
A6
A5
A4
A3
A2
A1
DIR
OE
CPU_D24
CPU_D25
CPU_D26
CPU_D27
CPU_D28
CPU_D29
CPU_D30
CPU_D31
RTC_D_SEL
CPU_RW
vINT1A

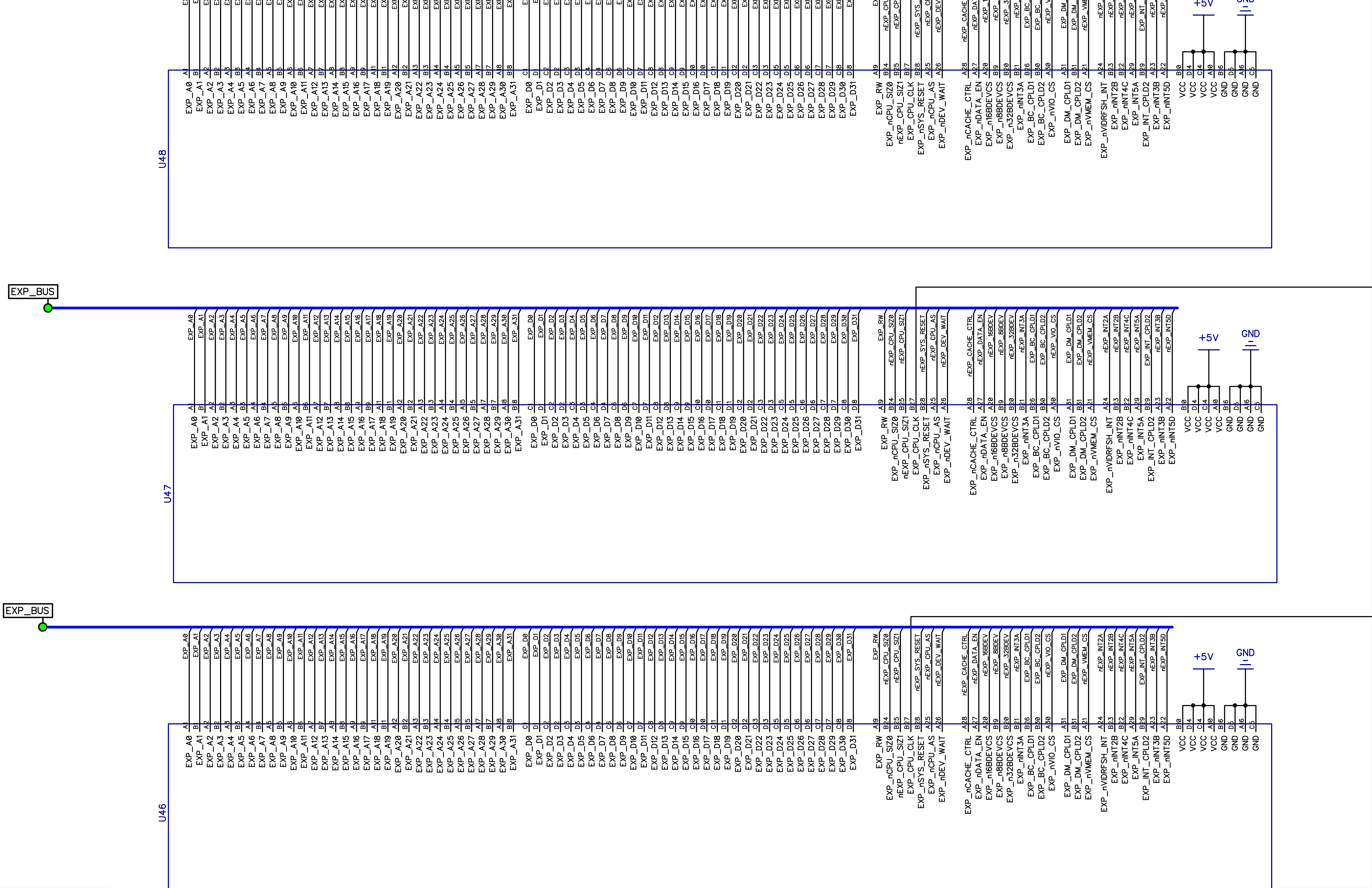


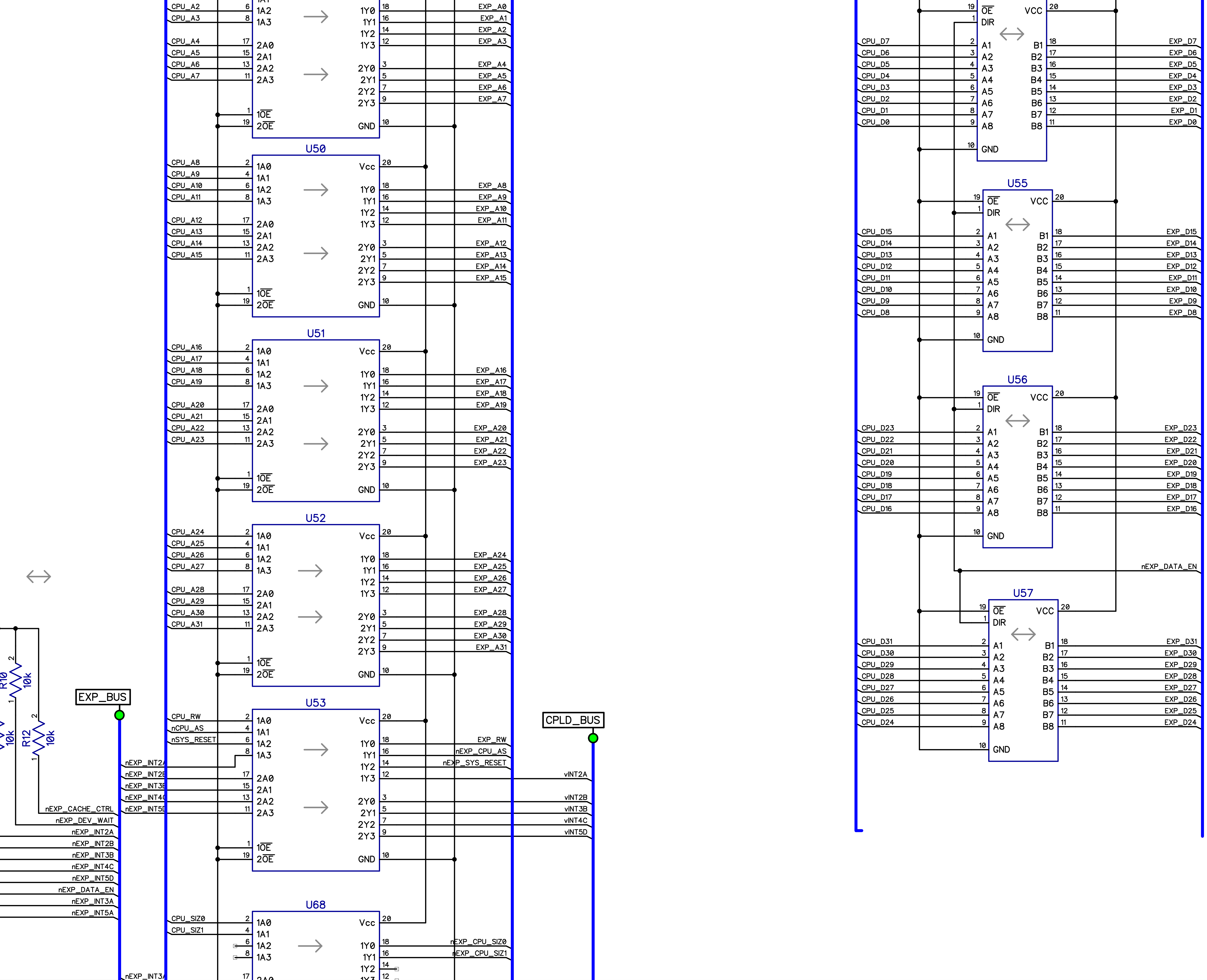
CPLD_BUS
CPU_BUS
+5V
GND
U36
ST16C2550
44PLCC
SYS_RESET_POS
CPU_A0
CPU_A1
CPU_A2
CPU_D24
CPU_D25
CPU_D26
CPU_D27
CPU_D28
CPU_D29
CPU_D30
CPU_D31
CPU_RW
nUART_CSB
nUART_CSA
nCPU_RD
vINT4B
vINT4A
GND
VCC
RESET
A0
A1
A2
D0
D1
D2
D3
D4
D5
D6
D7
/IOW
/CSB
/CSA
/IOR
INTB
INTA
/TXRDYA
/RXRDYA
/TXRDYB
/RXRDYB
XTAL1
XTAL2
RXA
TXA
/RTSA
/CTSA
/DTRA
/DSRA
/RIA
/CDA
/OP2A
RXB
TXB
/RTSB
/CTSB
/DTRB
/DSRB
/RIB
/CDB
/OP2B
R301
100k
J21
J301
UARTA
J20
UARTB
18.432MHz
U45
OE
OUT
VCC
GND
C16
R4
2k



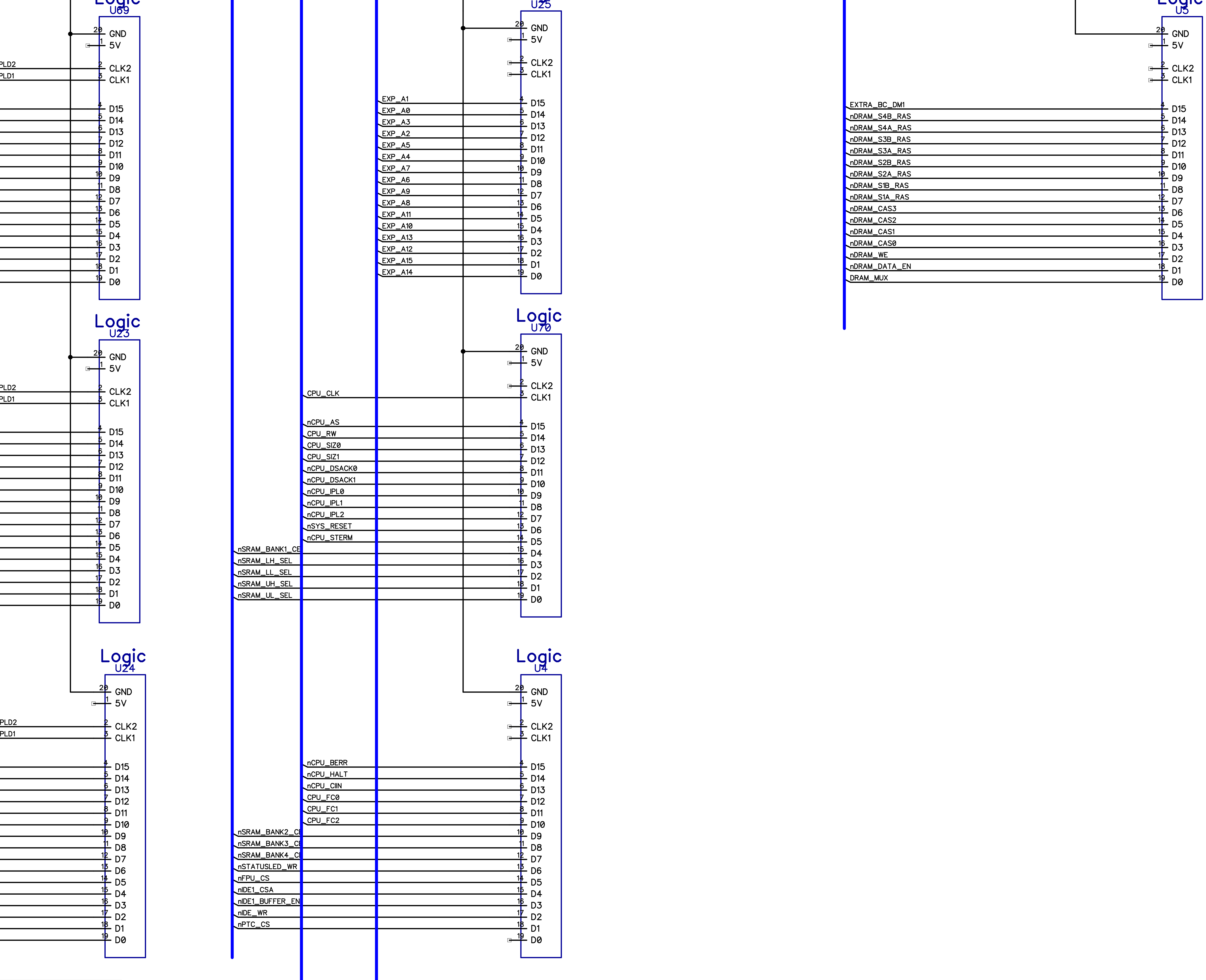
CPU_BUS
nIDE1_BUFFER_EN
CPU_RW
U38
U37
U39
J4
IDE Interface
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
A0
A1
A2
/IOW
/IOR
/CS0
/CS1
IRQR
IO_CH_RDY
KEY
/ACT
NC
/RESET
ALE
/IOCS16
GND
+5V
OE
DIR
VCC
CPU_D16
CPU_D17
CPU_D18
CPU_D19
CPU_D20
CPU_D21
CPU_D22
CPU_D23
CPU_D24
CPU_D25
CPU_D26
CPU_D27
CPU_D28
CPU_D29
CPU_D30
CPU_D31
CPU_A5
CPU_A6
CPU_A7
nIDE_WR
nCPU_RD
nIDE1_CSA
nIDE1_CSB
nSYS_RESET
vINT5B
U41
U40
U42
J6
J7

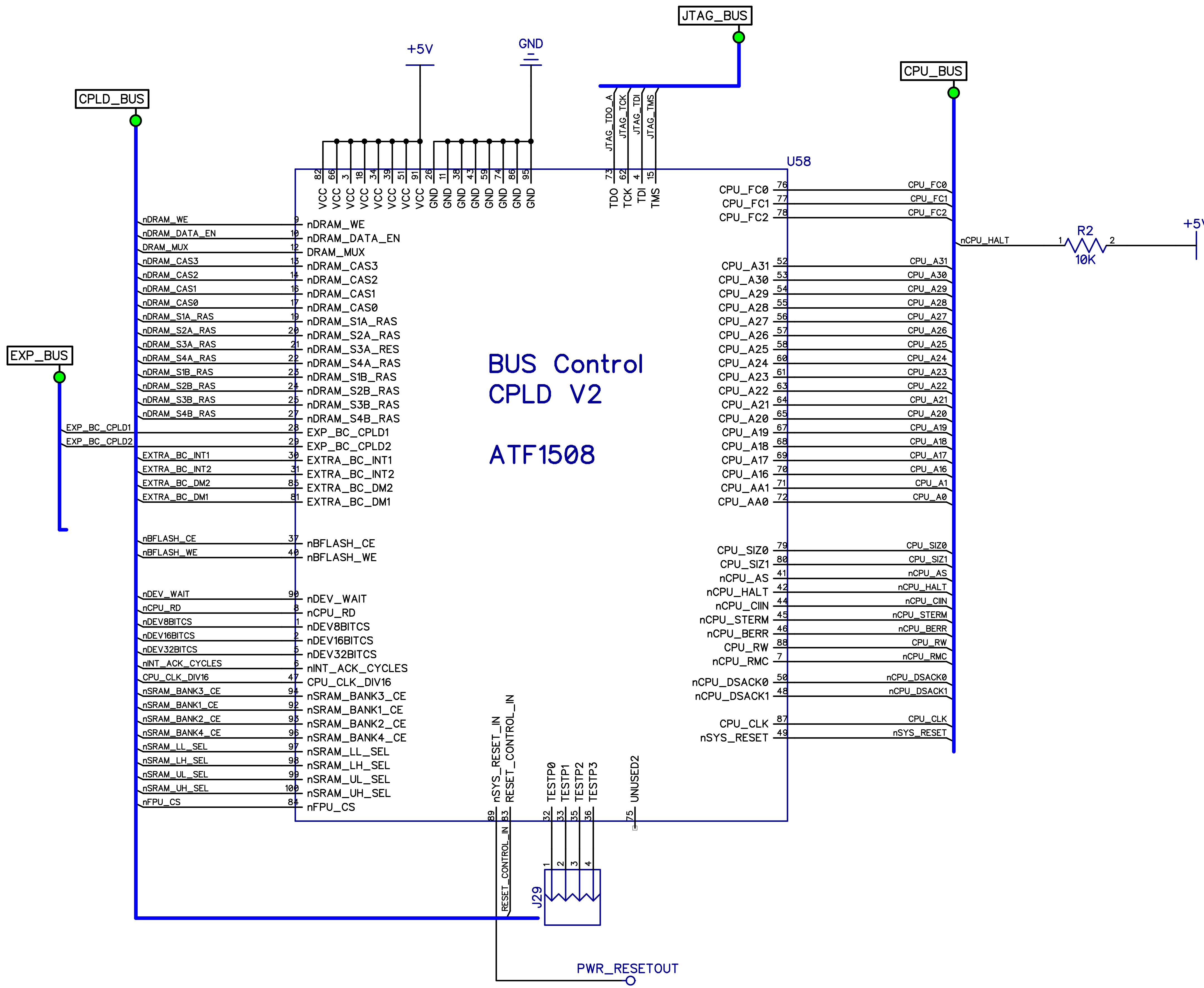

U64
ATX POWER
-12
+12V1
+12V1
+3V3
+3V3
+3V3
+3V3
NC
+5
+5VSB
/PS_ON
PWR_OK
GND
3V3
+5V
MAIN POWER
D5
R43
240
GND
R42
240
ALWAYS ON POWER
D4
J19
R41
10k
U65
VCC
GND
PSON
PWR_OK
AVR_MOSI
AVR_MISO
ACR_SCK
nAVR_RESET
RESET_BUTTON
POWER_BUTTON
POWER_LED_SINK
RESET_SINK
PWR_REQGR
SPARE
PWR_RESETOUT
To BUSCTRL CPLD to generate nSYS_RESET.
PWR_REQGR
To INTCTRL CPLD to indicate poweroff request AND to allow software poweroff.
R46
10k
J25
J24
U66
J17
U67
J18

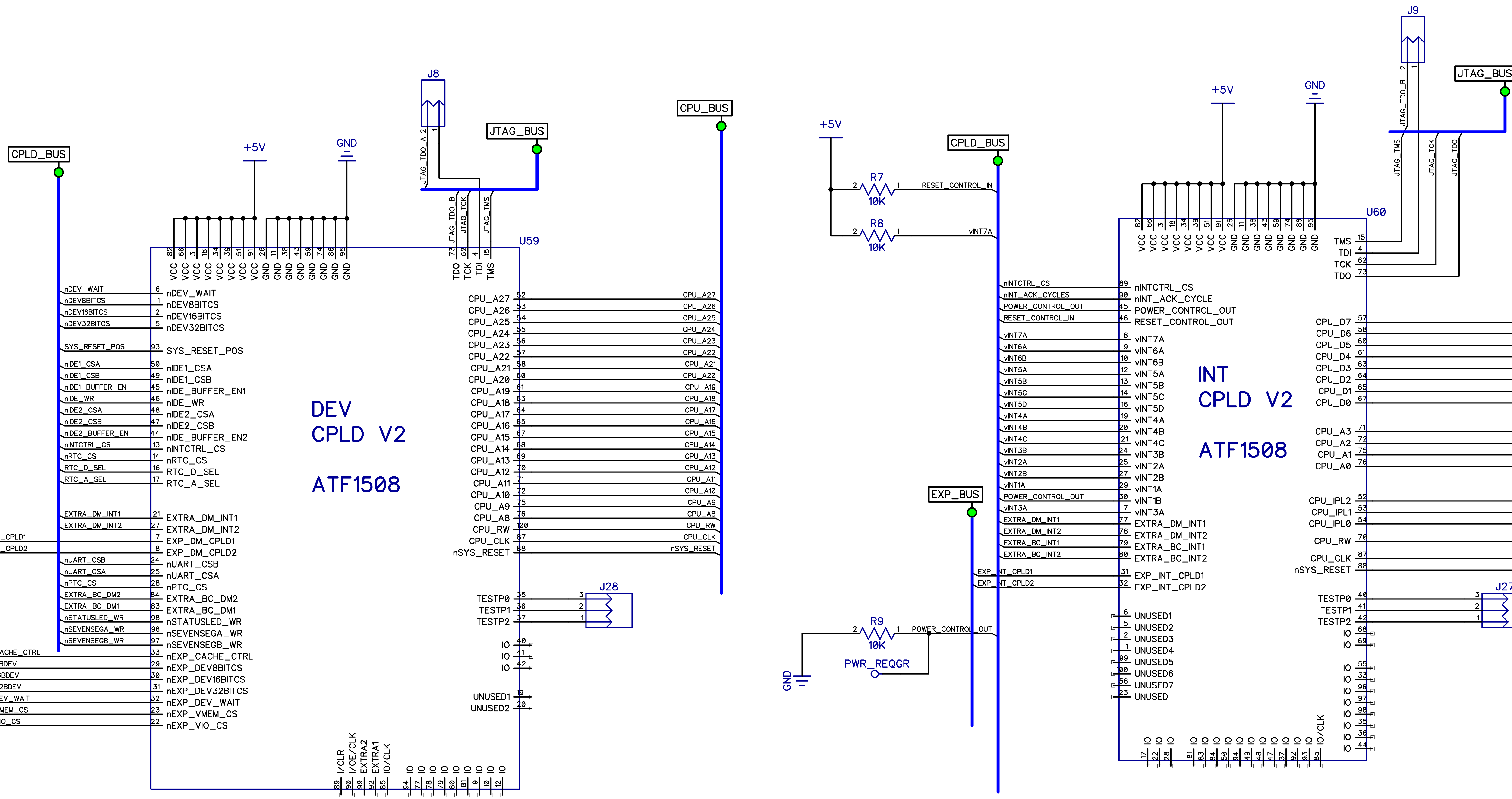

U48
EXP_BUS
U47
EXP_BUS
U46
+5V
GND
VCC




nEXP_DATA_EN:
0 = B->A EXP drives CPU
1 = A->B CPU drives EXP


Logic
U69
GND
5V
CLK2
CLK1
D15
D14
D13
D12
D11
D10
D9
D8
D7
D6
D5
D4
D3
D2
D1
D0
PLD2
PLD1
Logic
U23
Logic
U24
U25
EXP_A1
EXP_A0
EXP_A3
EXP_A2
EXP_A5
EXP_A4
EXP_A7
EXP_A6
EXP_A9
EXP_A8
EXP_A11
EXP_A10
EXP_A13
EXP_A12
EXP_A15
EXP_A14
Logic
U70
CPU_CLK
nCPU_AS
CPU_RW
CPU_SIZ0
CPU_SIZ1
nCPU_DSACK0
nCPU_DSACK1
nCPU_IPL0
nCPU_IPL1
nCPU_IPL2
nSYS_RESET
nCPU_STERM
nSRAM_BANK1_CE
nSRAM_LH_SEL
nSRAM_LL_SEL
nSRAM_UH_SEL
nSRAM_UL_SEL
Logic
U4
nCPU_BERR
nCPU_HALT
nCPU_CIIN
CPU_FC0
CPU_FC1
CPU_FC2
nSRAM_BANK2_CE
nSRAM_BANK3_CE
nSRAM_BANK4_CE
nSTATUSLED_WR
nFPU_CS
nIDE1_CSA
nIDE1_BUFFER_EN
nIDE_WR
nPTC_CS
Logic
U5
EXTRA_BC_DM1
nDRAM_S4B_RAS
nDRAM_S4A_RAS
nDRAM_S3B_RAS
nDRAM_S3A_RAS
nDRAM_S2B_RAS
nDRAM_S2A_RAS
nDRAM_S1B_RAS
nDRAM_S1A_RAS
nDRAM_CAS3
nDRAM_CAS2
nDRAM_CAS1
nDRAM_CAS0
nDRAM_WE
nDRAM_DATA_EN
DRAM_MUX


BUS Control
CPLD V2
ATF1508
U58
CPLD_BUS
EXP_BUS
JTAG_BUS
CPU_BUS
+5V
GND
R2
10K
J29
PWR_RESETOUT
VCC
GND
TDO
TCK
TDI
TMS
JTAG_TDO_A
JTAG_TCK
JTAG_TDI
JTAG_TMS
nDRAM_WE
nDRAM_DATA_EN
DRAM_MUX
nDRAM_CAS3
nDRAM_CAS2
nDRAM_CAS1
nDRAM_CAS0
nDRAM_S1A_RAS
nDRAM_S2A_RAS
nDRAM_S3A_RES
nDRAM_S4A_RAS
nDRAM_S1B_RAS
nDRAM_S2B_RAS
nDRAM_S3B_RAS
nDRAM_S4B_RAS
EXP_BC_CPLD1
EXP_BC_CPLD2
EXTRA_BC_INT1
EXTRA_BC_INT2
EXTRA_BC_DM2
EXTRA_BC_DM1
nBFLASH_CE
nBFLASH_WE
nDEV_WAIT
nCPU_RD
nDEV8BITCS
nDEV16BITCS
nDEV32BITCS
nINT_ACK_CYCLES
CPU_CLK_DIV16
nSRAM_BANK3_CE
nSRAM_BANK1_CE
nSRAM_BANK2_CE
nSRAM_BANK4_CE
nSRAM_LL_SEL
nSRAM_LH_SEL
nSRAM_UL_SEL
nSRAM_UH_SEL
nFPU_CS
nSYS_RESET_IN
RESET_CONTROL_IN
TESTP0
TESTP1
TESTP2
TESTP3
UNUSED2
CPU_FC0
CPU_FC1
CPU_FC2
nCPU_HALT
CPU_A31
CPU_A30
CPU_A29
CPU_A28
CPU_A27
CPU_A26
CPU_A25
CPU_A24
CPU_A23
CPU_A22
CPU_A21
CPU_A20
CPU_A19
CPU_A18
CPU_A17
CPU_A16
CPU_AA1
CPU_AA0
CPU_A1
CPU_A0
CPU_SIZ0
CPU_SIZ1
nCPU_AS
nCPU_CIIN
nCPU_STERM
nCPU_BERR
CPU_RW
nCPU_RMC
nCPU_DSACK0
nCPU_DSACK1
CPU_CLK
nSYS_RESET



CPLD_BUS
+5V
GND
J8
JTAG_BUS
CPU_BUS
U59
DEV
CPLD V2
ATF1508
VCC
GND
TDO
TCK
TDI
TMS
JTAG_TDO_A
JTAG_TDO_B
JTAG_TCK
JTAG_TMS
nDEV_WAIT
nDEV8BITCS
nDEV16BITCS
nDEV32BITCS
SYS_RESET_POS
nIDE1_CSA
nIDE1_CSB
nIDE_BUFFER_EN1
nIDE_WR
nIDE2_CSA
nIDE2_CSB
nIDE_BUFFER_EN2
nINTCTRL_CS
nRTC_CS
RTC_D_SEL
RTC_A_SEL
EXTRA_DM_INT1
EXTRA_DM_INT2
EXP_DM_CPLD1
EXP_DM_CPLD2
nUART_CSB
nUART_CSA
nPTC_CS
EXTRA_BC_DM2
EXTRA_BC_DM1
nSTATUSLED_WR
nSEVENSEGA_WR
nSEVENSEGB_WR
nEXP_CACHE_CTRL
nEXP_DEV8BITCS
nEXP_DEV16BITCS
nEXP_DEV32BITCS
nEXP_DEV_WAIT
nEXP_VMEM_CS
nEXP_VIO_CS
CPU_A27
CPU_A26
CPU_A25
CPU_A24
CPU_A23
CPU_A22
CPU_A21
CPU_A20
CPU_A19
CPU_A18
CPU_A17
CPU_A16
CPU_A15
CPU_A14
CPU_A13
CPU_A12
CPU_A11
CPU_A10
CPU_A9
CPU_A8
CPU_RW
CPU_CLK
nSYS_RESET
TESTP0
TESTP1
TESTP2
J28
IO
UNUSED1
UNUSED2
I/CLR
I/OE/CLK
EXTRA2
EXTRA1
IO/CLK
R7
10K
RESET_CONTROL_IN
R8
vINT7A
CPLD_BUS
EXP_BUS
R9
POWER_CONTROL_OUT
PWR_REQGR
J9
U60
INT
CPLD V2
ATF1508
nINTCTRL_CS
nINT_ACK_CYCLE
POWER_CONTROL_OUT
RESET_CONTROL_OUT
vINT7A
vINT6A
vINT6B
vINT5A
vINT5B
vINT5C
vINT5D
vINT4A
vINT4B
vINT4C
vINT3B
vINT2A
vINT2B
vINT1A
vINT1B
vINT3A
EXTRA_DM_INT1
EXTRA_DM_INT2
EXTRA_BC_INT1
EXTRA_BC_INT2
EXP_INT_CPLD1
EXP_INT_CPLD2
UNUSED1
UNUSED2
UNUSED3
UNUSED4
UNUSED5
UNUSED6
UNUSED7
UNUSED
CPU_D7
CPU_D6
CPU_D5
CPU_D4
CPU_D3
CPU_D2
CPU_D1
CPU_D0
CPU_A3
CPU_A2
CPU_A1
CPU_A0
CPU_IPL2
CPU_IPL1
CPU_IPL0
CPU_RW
CPU_CLK
nSYS_RESET
TESTP0
TESTP1
TESTP2
J27